

Subsets of $\mathbb{F}_p$ with two prescribed moments

Part I: the pair $\{1, 3\}$ — a moment subset sum over $\mathbb{F}_p$: exact counts of totally split squarefree polynomials with a non-consecutive pair of prescribed coefficients, and exactly when those counts are elementary

Carles Marín*

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Abstract

When the non-consecutive coefficients of X^{n-1} and X^{n-3} are both prescribed to be zero, the exact count of totally split squarefree polynomials of degree $n \geq 3$ over $\mathbb{F}_p$ admits an *elementary* expression — a polynomial in p with coefficients in quadratic characters of fixed integers — for exactly three degrees, $n = 3, 4, 6$. This note proves that. It also gives four exact closed forms, one of them for $n = 5$, where the expression is explicit but involves a Frobenius trace and is therefore not elementary in that sense: an exact formula and an elementary one are different things, and it is the second that is classified here. Fix an odd prime p and let $C(n, p)$ count the n -subsets $T \subseteq \mathbb{F}_p$ with $\sum_{t \in T} t = \sum_{t \in T} t^3 = 0$; equivalently, for $p > 3$, the monic degree- n polynomials over $\mathbb{F}_p$ that are totally split and squarefree and whose coefficients of X^{n-1} and X^{n-3} both vanish. This is the quantity written $C_{\{1,3\}}(n, p)$ in the S -matrix note [20], where its main term and an explicit error bound are given and where a closed form is left open. We evaluate it: $C(3, p)$, $C(4, p)$, $C(5, p)$ and $C(6, p)$ are given in closed form, the case $n = 5$ involving the Frobenius trace of the elliptic curve of conductor 20. We then answer the question that those four identities pose, namely for which n an *elementary* expression can exist at all. Call $C(n, p)$ *elementary* if it is a polynomial in p whose coefficients involve only quadratic characters of fixed integers. Our main result is that for $n \geq 3$, $C(n, p)$ is elementary precisely for $n = 3, 4, 6$. The proof is short and the mechanism is combinatorial: the partition sieve attaches to each $\lambda \vdash n$ a weighted hyperplane section of a diagonal cubic, whose singular points are the sign vectors splitting the weights into two blocks of equal sum — the *biquanimous* splittings of λ — and it is the stratum of length exactly four that decides, because there the variety is a plane cubic and smoothness forces an elliptic curve. The uniform witness $\lambda = (n-3, 1, 1, 1)$ settles every $n > 6$ at once, since then $n-3 > 3$ and no equal-sum splitting exists; $n = 5$ follows because its total is odd.

Two further strata are worked out in full, by pencils of conics, because they show how far the length-five band is from being uniform: $V_{(a,1,1,1,1)}$ has line field $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$, abelian of exponent two for every a , while $V_{(2,2,1,1,1)}$ — which also has exactly two distinct weights — does not, its line field containing the splitting field of a cubic of discriminant -140 and Galois group S_3 that the same pencil produces rather than quotes. Finally, the two extremal cubics that the sieve throws up are identified with their classical models *over the base field* and not merely over the closure, by exhibiting the map: the four rational nodes of $V_{(2,1^4)}$ and the matrix carrying it to Cayley’s standard form, and $V_{(1^6)}$ as the Segre cubic on the nose.

Keywords: moment subset sum; subset sum over finite fields; finite fields; prescribed coefficients; Li–Wan sieve; diagonal cubics; Segre cubic; Cayley nodal cubic; Clebsch diagonal surface; Waring rank; pentahedral form; biquanimous partitions; elliptic curves.

1 The question, and whose it is

Let p be an odd prime. For a set J of positive integers with $1 \in J$, the S -matrix note [20] writes $C_J(n, p)$ for the number of n -subsets $T \subseteq \mathbb{F}_p$ with $M_j(T) = \sum_{t \in T} t^j = 0$ for every $j \in J$, and

*Independent researcher, karlesmarin@gmail.com. Written with Claude (Anthropic) as an assistant.

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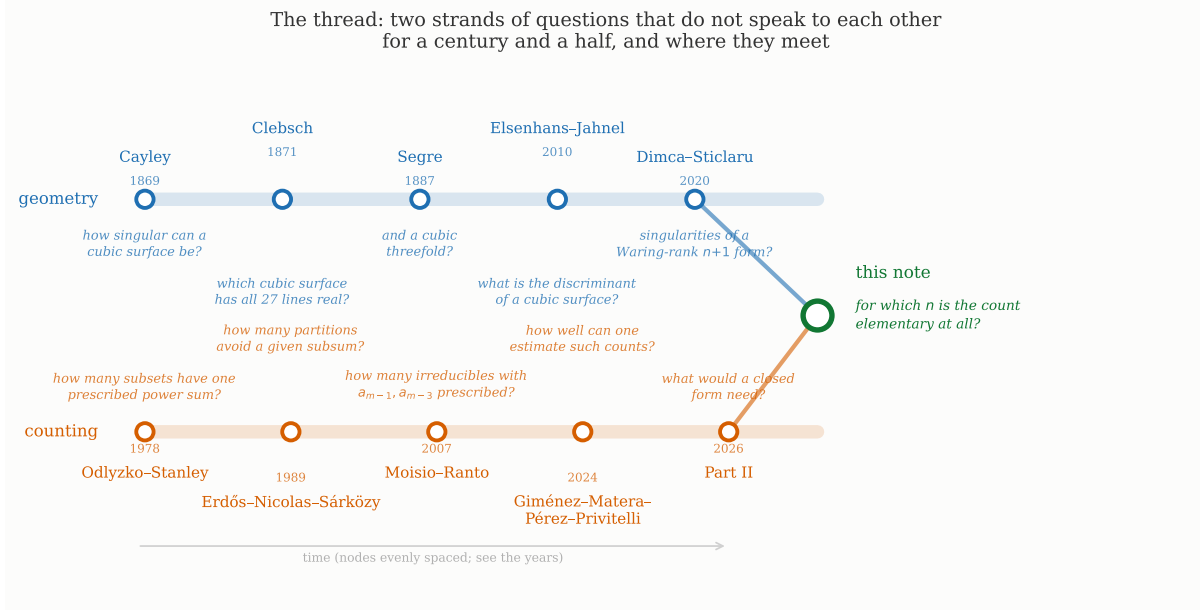


Figure 1: Two strands of questions. The geometric one asks how singular a cubic can be; the arithmetic one asks how many polynomials there are with prescribed coefficients. They run for a century and a half without meeting. The question left open at the end of the arithmetic strand — what would a closed form need? — is answered here, and so is the sharper one it opens: for which n the count is *elementary* at all. That second answer needs an object from the other strand. Nodes are evenly spaced, and the years label *papers*: Cayley’s memoir on cubic surfaces is [28] and Clebsch’s is [29], so the surface is dated by its memoir and not by the plaster model of the following year that Appendix A.3 describes.

proves there the estimate

$$C_J(n, p) = \frac{1}{p^{|J|}} \binom{p}{n} + O_n(p^{n/2}) \quad (1)$$

together with an explicit bound on the error. That paper needs only the estimate, and it closes its list of open problems with the observation that what the available estimates do not give is a closed form, and that

“it is those weights, together with the singular locus itself rather than a bound on its dimension, that a closed form seems to need.”

This note takes $J = \{1, 3\}$ — a *non-consecutive* pair — writes $C(n, p) = C_{\{1,3\}}(n, p)$ for brevity, and does exactly that.

Nothing in (1) is reproved here. The object, its notation, its main term, its error bound and the inclusion–exclusion identity relating $C_{\{1,2\}}$, $C_{\{1,3\}}$ and $C_{\{1,2,3\}}$ all belong to [20]. What is new is the evaluation.

What is in the literature already

The enumeration of polynomials over a finite field with prescribed coefficients is old and well populated, and the reader should have the following in view.

Giménez, Matera, Pérez and Privitelli [7] allow an *arbitrary* index set $I = \{i_1 < \dots < i_m\}$ of prescribed coefficients subject to $i_m \leq n - 3$ — which covers our $\{1, 3\}$ as soon as $n \geq 6$ — and bound the count for each factorisation pattern. Gottig, Pérez and Privitelli [9] reach the same

question through systems of diagonal equations; Lai, Marino, Robinson and Wan [13] through the sieve; and Odlyzko and Stanley [18] gave, long before, a closed form for subsets of $\mathbb{F}_p^\times$ of unrestricted size with a single prescribed power sum. All of these are estimates, or concern a single moment. Kuttner and Wang [10] and Gao [6] build exact generating-function frameworks for prescribed *leading* coefficients. In that language our pair is three leading coefficients with the middle one left free, whereas the evaluated results of [10] are for at most one prescribed leading coefficient, and those of [6] require a ground set whose size is an even power of the characteristic; so neither evaluates the specialisation treated here. The evaluated term of [10], summed over the free coefficient, is in our regime the main term $p^{-|J|}\binom{p}{n}$ of (1); the structure lives in the difference.

The consecutive pair $\{1, 2\}$ with both moments zero is evaluated in closed form, for every subset size, by Nguyen [17, Thm. 2.11] and by Zhang, Zheng, Wang and Yuan [16, Cor. 21]. There the two conditions are a congruence $\prod_i (1 + x_i y) \equiv 1 \pmod{y^3}$ in a truncated unit group, whose characters lead to quadratic Gauss sums of known value. The non-consecutive pair loses that shape: $e_1 = e_3 = 0$ leaves e_2 free, the strata of length four become plane cubics, and a Frobenius trace can survive in the count.

Closest in spirit is Moisiso and Ranto [15], who count *irreducible* polynomials with a_{m-1} and a_{m-3} prescribed — the same index pair as ours — and find that the problem is governed by rational point counts on fibre products of supersingular elliptic curves and Kloosterman curves. Their work is in characteristic 2 and concerns the opposite factorisation pattern; but it establishes, in 2008, that the pair $\{1, 3\}$ is a pattern for which elliptic curves appear, and their Lemma 4 is the same Newton computation that we record below as Lemma 1. Taken together with what follows, the index pair $\{1, 3\}$ produces elliptic curves at *both* ends of the factorisation spectrum — irreducible in characteristic two, totally split in odd characteristic — by routes that do not meet. We have no explanation for that, and offer it as an observation.

2 From coefficients to moments, and the sieve

Lemma 1. *Let $p > 3$ and let $T \subseteq \mathbb{F}_p$ with $|T| = n$. Then $M_1(T) = M_3(T) = 0$ if and only if the coefficients of X^{n-1} and X^{n-3} in $\prod_{t \in T} (X - t)$ both vanish.*

Proof. Write e_j for the elementary symmetric functions of T . Newton's identity gives $M_3 = e_1 M_2 - e_2 M_1 + 3e_3$, so $e_1 = M_1 = 0$ forces $M_3 = 3e_3$, and 3 is invertible. $\square$

Lemma 1

Newton's identity, read off

Nothing here is ours: it is one line of Newton's identities, recorded because the whole note works with $e_1 = e_3 = 0$ rather than with $M_1 = M_3 = 0$, and the translation between the two needs $p > 3$. That hypothesis is the only content, and it is why $p = 3$ is excluded throughout.

The hypothesis $p > 3$ is not cosmetic. In characteristic 3 one has $M_3 = M_1^3$, so M_3 vanishes identically once M_1 does and says nothing about e_3 ; in characteristic 2 the factor 3 is 1 and the conversion is immediate, which is the form in which the computation appears in [15, Lemma 4].

So $C(n, p)$ counts the monic degree- n polynomials over $\mathbb{F}_p$ that are totally split and squarefree and lie in the linear family

$$X^n + 0 \cdot X^{n-1} + a_2 X^{n-2} + 0 \cdot X^{n-3} + a_4 X^{n-4} + \dots$$

of dimension $n - 2$.

A partition sieve of Li–Wan type [14, 13] converts this into a count without the distinctness condition. For $\lambda = (w_1, \dots, w_l) \vdash n$ put $z_\lambda = \prod_j j^{m_j} m_j!$, where m_j is the number of parts equal to j . Then

$$C(n, p) = \sum_{\lambda \vdash n} \frac{(-1)^{n-l(\lambda)}}{z_\lambda} N_\lambda, \quad N_\lambda = \#\left\{x \in \mathbb{F}_p^l : \sum_i w_i x_i = 0, \sum_i w_i x_i^3 = 0\right\}. \quad (2)$$

The system defining N_λ is a cone, so $N_\lambda = 1 + (p - 1) \#V_\lambda(\mathbb{F}_p)$ with

$$V_\lambda = \left\{ \sum_i w_i x_i = 0 \right\} \cap \left\{ \sum_i w_i x_i^3 = 0 \right\} \subset \mathbb{P}^{l-1}, \quad (3)$$

a weighted hyperplane cutting a diagonal cubic *with the same weights*. For $l \geq 4$ the two equations meet properly and $\dim V_\lambda = l - 3$: the stratum $l = 4$ gives plane cubics, $l = 5$ cubic surfaces, $l = 6$ cubic threefolds. For $l \leq 3$ that count fails and the strata are handled directly: $\lambda = (3, 3)$, for instance, has $x + y = 0$ and $x^3 + y^3 = 0$ with the second equation identically zero on the first, so $V_{(3,3)} = \{(1 : -1)\} \subset \mathbb{P}^1$ is a single reduced smooth point rather than a variety of dimension $l - 3 = -1$. Tables 1 and 2 record the strata with $l \leq 2$ by their N_λ , which is what the sieve uses, and those with $l = 3$ by $\#V_\lambda(\mathbb{F}_p)$, which determines it.

Measured: (2) against the exact census, $n \leq 7$ and $p \leq 53$ — every $3 \leq n \leq 7$ and every odd prime $p \leq 53$, 75 cases — with no exception, the N_λ computed by dynamic programming over the two sums rather than read from the tables. See `P3c.py`.

3 The singular locus, and the count it produces

3.1 The lemma, and whose it is

Lemma 2 (Segre; Elsenhans–Jahnel; Dimca–Sticlaru). *Let k be an algebraically closed field of characteristic $\neq 2, 3$, let $l \geq 4$, let $w_1, \dots, w_l \in k^\times$ be arbitrary, and let V_w be as in (3). Then*

$$\text{Sing}(V_w) = \left\{ [\varepsilon] : \varepsilon_i = \pm 1, \sum_{i:\varepsilon_i=+1} w_i = \sum_{i:\varepsilon_i=-1} w_i \right\}$$

up to a global sign, and every such point is an ordinary double point.

Proof. A point of the complete intersection is singular exactly when the two gradients are proportional, i.e. when $3w_i x_i^2 = c w_i$ for some c and all i . As $w_i \neq 0$ this says $x_i^2 = c/3$ for every i ; $c = 0$ gives only the origin, and for $c \neq 0$ we may rescale so that $x_i = \pm 1$. The cubic equation then holds automatically, since $(\pm 1)^3 = \pm 1$, and the linear one says exactly that the weights split into two blocks of equal sum. For the node: at such an ε the quadratic term is $Q(v) = \sum_i w_i \varepsilon_i v_i^2$ restricted to $L(v) = \sum_i w_i v_i = 0$. Now Q is nondegenerate on the ambient space and the Q -orthogonal of $\ker L$ is the line $k\varepsilon$, which lies inside $\ker L$ precisely by the balance condition; so the radical of the restriction is exactly $k\varepsilon$, and the form descends nondegenerately to the projective tangent space. $\square$

We claim no part of Lemma 2. With all weights 1 and $l = 6$ the variety is the Segre cubic and the ten points are its classical nodes. In the pentahedral normal form obtained by setting $y_i = w_i x_i$ — which turns (3) into $\sum y_i = 0$, $\sum y_i^3 / w_i^2 = 0$, a rational Sylvester pentahedral form with coefficients $a_i = w_i^{-2}$ — the statement for $l = 5$, the case of cubic surfaces, is exactly [3, Fact 2.8], whose Examples 2.9 are three instances of our V_λ with their node counts recorded. For arbitrary degree and arbitrary coefficients it is [2, Theorem 2.4].

Lemma 2

not ours — cited

Base case Segre [30]. Stated for surfaces, in the pentahedral coordinates used here, by Elsenhans–Jahnel [3, Fact 2.8], whose Examples 2.9 are three of our V_λ with their node counts printed. For arbitrary degree and coefficients, Dimca–Sticlaru [2, Thm. 2.4]. The proof is given because the note needs it explicit, not because anything is claimed.

3.2 The count

Definition 3. A multiset of positive integers is *biquanimous* if it can be split into two sub-multisets of equal total. The number of partitions of $2n$ that are *not* biquanimous is OEIS A006827; their asymptotics are due to Erdős, Nicolas and Sárközy [5].

Corollary 4. For $\lambda \vdash n$ of length $l \geq 4$ and $p > n$,

$$\# \text{Sing}(V_\lambda) = \frac{1}{2} \# \{A \subseteq [l] : \sum_{i \in A} w_i = n/2\}.$$

In particular, for such λ , V_λ is smooth if and only if λ is not biquanimous, and for odd n every V_λ of length at least four is smooth.

The restriction $l \geq 4$ is not cosmetic. For $\lambda = (3, 3)$ and $p > 6$ the two equations are $x + y = 0$ and $x^3 + y^3 = 0$, the second identically zero on the first, so $V_{(3,3)} = \{(1 : -1)\}$ is a single reduced smooth point and its singular locus is empty — whereas the right-hand side above counts 1, because $(3, 3)$ splits into two blocks of equal sum. The short strata are handled directly in Tables 1 and 2 — by N_λ for $l \leq 2$ and by $\#V_\lambda(\mathbb{F}_p)$ for $l = 3$ — and never by this corollary.

Corollary 4

reading of Lemma 2, not a new result

It is Lemma 2 counted: the sign vectors ε with $\sum_i w_i \varepsilon_i = 0$ are in two-to-one correspondence with the equal-sum splittings, since ε and $-\varepsilon$ give the same point. What is ours is only the observation that this turns the singular locus into a partition-counting problem and therefore into OEIS A006827 [5]. The hypothesis $l \geq 4$ is inherited from the lemma and must be carried: without it the statement is false, as $(3, 3)$ shows.

This is immediate from Lemma 2, and we record it only because it is the form in which the lemma will be used: it replaces a question about a variety by a subset-sum question about λ , and it makes the singular points visibly $\mathbb{F}_p$ -rational, which is what the blow-up models of the Segre and Cayley strata in Section 5 rest on as a geometric reading; the counts there are made directly. All of this assumes $p > n$, which in particular makes every weight invertible; that hypothesis cannot be dropped on its own. For $\lambda = (5, 1, 1, 1)$ and $p = 5$ the weight 5 vanishes, the equations collapse to $x_2 + x_3 + x_4 = 0$ and $x_2^3 + x_3^3 + x_4^3 = 0$ with x_1 free, and eliminating x_4 gives $-3x_2x_3(x_2 + x_3) = 0$: three concurrent lines whose common point $(1 : 0 : 0)$ is not a sign vector and not an ordinary double point. Outside $p > n$ one needs at least $p \nmid w_1 \cdots w_l$ and $p \neq 2, 3$. Over $\mathbb{F}_p$ the correct condition is then the congruence $2 \sum_A w \equiv n \pmod{p}$, which agrees with the equality exactly when $p > n$; for $p \leq n$ there can be extra points, and not only when $p \mid n$: for $\lambda = (4, 1, 1, 1)$ and $p = 5$ the three sign vectors with a single -1 on a weight-one coordinate are singular, although $(4, 1, 1, 1)$ is not biquanimous.

Measured: Corollary 4 against brute force over $\mathbb{F}_p$, $7 \leq p \leq 23$, in the cases it covers, $n \leq 7$ and $4 \leq l \leq 5$. The 128 cases run span $3 \leq l \leq 5$; the 66 with $l = 3$ lie *outside* the corollary and agree with the sign-vector count all the same, so they are not evidence that the hypothesis $l \geq 4$ is needed. The case that needs it is $(3, 3)$, of length two, treated above: the ideal $(x + y, x^3 + y^3)$ is $(x + y)$, a reduced point, while the sign-vector count gives 1. Nine of the 128 have $p \mid n$, where the congruence form correctly predicts the extra points. See `gates/P3b_singularidades.py`.

4 When the count is elementary

Definition 5. Let $n \geq 3$. We say $C(n, p)$ is *elementary* if there are a polynomial $P \in \mathbb{Q}[T, U_1, \dots, U_r]$ and fixed integers $d_1, \dots, d_r$ such that $C(n, p) = P(p, \chi_p(d_1), \dots, \chi_p(d_r))$ for all sufficiently large p , where χ_p is the quadratic character. The coefficients may be rational, not only integral: $C(3, p) = (p - 1)/2$ already has a denominator, and an integrality requirement would exclude it. Integrality of the *values* of $C(n, p)$ is a different matter. The restriction $n \geq 3$ is needed for the statement to be about the coefficient of X^{n-3} at all, and it also removes the degenerate elementary cases $C(1, p) = 1$ and $C(2, p) = (p - 1)/2$.

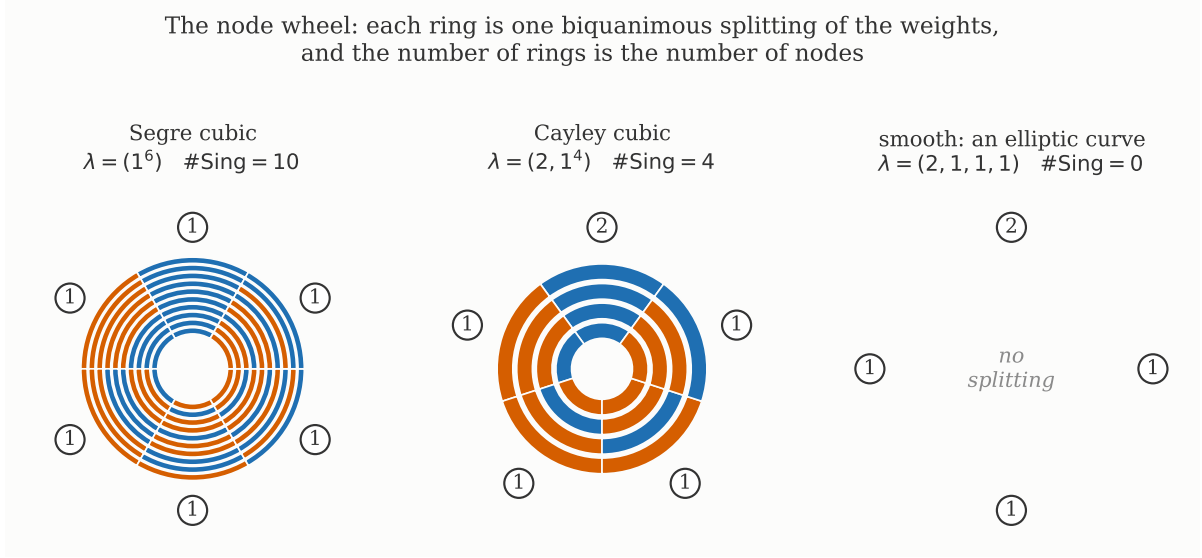


Figure 2: The same statement, drawn. Each weight is a bead; each ring is one way of two-colouring the beads so that the two colours carry equal total weight. The number of rings is $\# \text{Sing}$. The third wheel has no ring, which is what smoothness looks like.

Theorem 6. $C(n, p)$ is elementary if and only if every $\lambda \vdash n$ with exactly four parts is biquanimous, and this happens precisely for

$$n \in \{3, 4, 6\}.$$

Proof. For $n = 3, 4, 6$ the closed forms of Section 5 are elementary, and there is nothing to prove.

For the converse, let $n \geq 5$ with $n \neq 6$ and take

$$\lambda^* = (n - 3, 1, 1, 1).$$

That λ^* is not biquanimous needs the two cases separated. For $n > 6$ a sub-multiset omitting the large part has total at most $3 < n/2$, while one containing it has total at least $n - 3 > n/2$, so neither can equal $n/2$; this covers every $n > 6$, odd or even. It does not cover $n = 5$, where both inequalities fail, and there the reason is parity: $n/2$ is not an integer, so no sub-multiset can total it. Hence λ^* is not biquanimous for every $n \geq 5$ with $n \neq 6$ and, by Corollary 4, V_{λ^*} is smooth. As $l(\lambda^*) = 4$ it is a smooth plane cubic, hence a curve of genus one, and $\#V_{\lambda^*}(\mathbb{F}_p) = p + 1 - a_p$ for the Frobenius trace a_p of its Jacobian.

It remains to see that such a stratum cannot be cancelled. Fix an auxiliary prime ℓ and work in the rationalised Grothendieck group of ℓ -adic Galois representations, writing $e(\cdot)$ for the compactly supported class and $\mathbb{L}$ for the Lefschetz motive, and write C_λ for the affine cone over V_λ , whose $\mathbb{F}_p$ -points N_λ counts, so that $e(C_\lambda) = 1 + (\mathbb{L} - 1)e(V_\lambda)$. For a smooth stratum of length four, $e(V_\lambda) = 1 - H_\lambda + \mathbb{L}$ with $H_\lambda = H^1$ of the Jacobian, so

$$e(C_\lambda) = \mathbb{L}^2 + (1 - \mathbb{L})H_\lambda,$$

whose weight-one part is $+H_\lambda$.

A *singular* stratum of length four carries no weight one either, and this is the step on which the whole criterion rests. By Lemma 2 its singularities are ordinary double points, and the statement we need is about components, not about rationality of the whole curve: *every geometric component of the normalisation of a reduced singular plane cubic has genus zero*. For

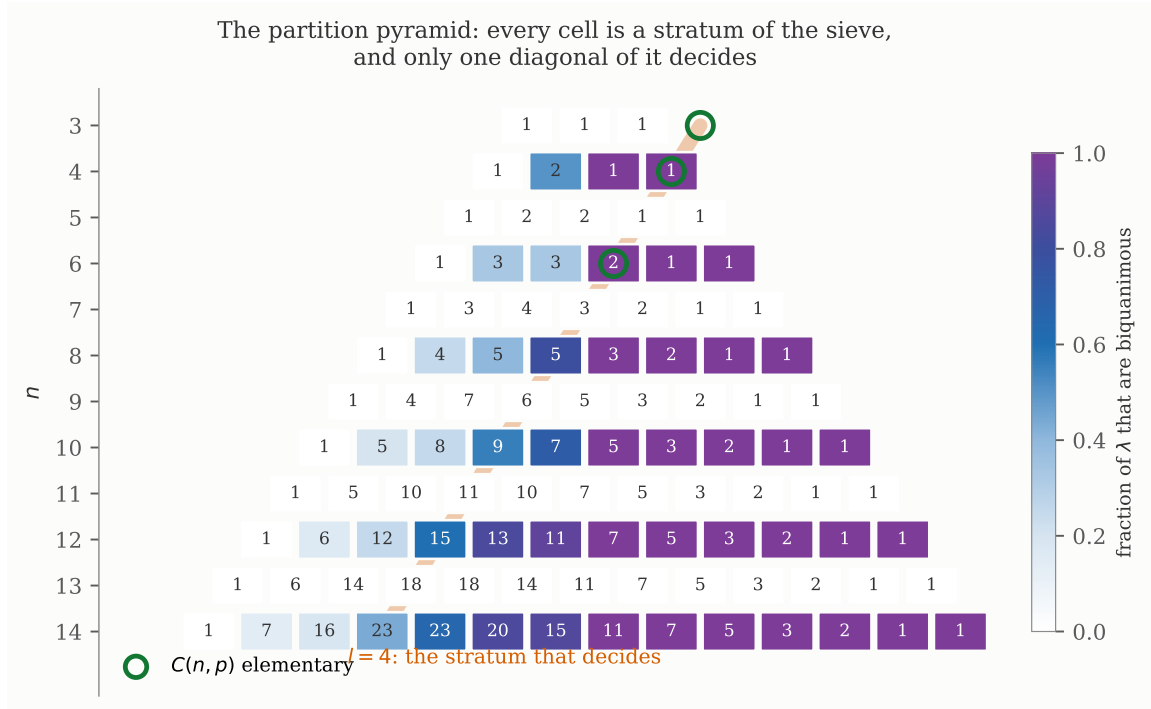


Figure 3: The sieve runs over the partitions of n by length, which is a triangular array. Shade is the fraction of λ in that cell which are biquanamous; odd rows are blank because an odd total cannot be halved. The theorem says that only the marked diagonal decides, and the circled rows are the three where it is empty or entirely biquanamous.

an irreducible nodal cubic that is the usual rational parametrisation; for a reducible one — and these occur here, the stratum $(2^2, 1^2)$ of $n = 6$ being a line and a conic, with count $2p$ — each component is a line or a conic, so again genus zero. In both cases H^1 of the normalisation vanishes, and the corrections coming from identifying points on it are supported on finite sets and are Artin of weight zero.

It is worth saying explicitly that this is special to degree three. The right statement is not that a node cannot destroy weight one in higher degree: a plane quartic with three nodes has geometric genus $\binom{3}{2} - 3 = 0$ and carries none. It is that in degree $d \geq 4$ singularity *alone* no longer suffices — one node lowers the genus by one and leaves $\binom{d-1}{2} - 1$, which is positive for $d \geq 4$, so the implication needs to know how many singularities there are. The condition that does the work in general is that the normalisation have genus zero, and it coincides with mere singularity only in degree three.

Strata with $l \leq 3$ are zero-dimensional or unions of rational curves and carry no weight one. For $l \geq 5$ the variety V_λ is cut in $\mathbf{P}^{l-1}$ by $r = 2$ forms, so Ghorpade–Lachaud [8, Prop. 3.2] makes $H^i(\mathbf{P}^{l-1}) \rightarrow H^i(V_\lambda)$ an isomorphism for $i \leq l - 1 - 2 - 1 = l - 4$; taking $i = 1$ needs exactly $l \geq 5$ and gives $H^1(V_\lambda) = 0$. That is a statement about V_λ and not about its resolution, which is no longer the complete intersection we started from, so one further step is needed. Let $f : \tilde{V}_\lambda \rightarrow V_\lambda$ blow up the nodes. Each exceptional fibre is a smooth quadric of dimension $l - 4 \geq 1$, geometrically connected and with vanishing H^1 , so proper base change gives $f_* \mathbb{Q}_\ell = \mathbb{Q}_\ell$ and $R^1 f_* \mathbb{Q}_\ell = 0$, and the low-degree Leray sequence yields $H^1(\tilde{V}_\lambda) \simeq H^1(V_\lambda) = 0$. Since $\tilde{V}_\lambda$ is smooth and proper, weight one can only come from H^1 , which vanishes; and in

$$e(V_\lambda) = e(\tilde{V}_\lambda) - \sum_s e(Q_s) + \sum_s e(s)$$

the quadrics and the points contribute only even weights. So those strata carry no weight one either. This does not say that all odd weights vanish on the long strata — only weight one,

which is what the argument uses. Finally, the sieve sign $(-1)^{n-l}$ is, on the stratum $l = 4$, equal to $(-1)^{n-4}$ and hence *independent of λ* . Writing R_n for the virtual sieve class,

$$\mathrm{gr}_1^W R_n = (-1)^{n-4} \sum_{\substack{\lambda \vdash n, l(\lambda)=4 \\ V_\lambda \text{ smooth}}} \frac{[H_\lambda]}{z_\lambda},$$

where the *sum* is a positive rational combination of two-dimensional representations. The global sign $(-1)^{n-4}$ is negative for n odd — already for $n = 5$, the first case the witness covers — so the displayed class is not positive; what the argument uses is only that it is *nonzero*, which the positivity of the sum gives as soon as one such λ exists.

An elementary expression, being a polynomial in p with character coefficients, is the Frobenius trace of a virtual representation with only *even* weights. To compare the two, work with the ℓ -adic representations of $G_{\mathbb{Q}}$ attached to the varieties V_λ , which are defined over $\mathbb{Q}$ by the equations (3) with integer weights, discarding the finite set of primes of bad reduction and those dividing the weights, and multiply both virtual classes by a common denominator so that the rational multiplicities become integral. Then by Chebotarev and Brauer–Nesbitt, equality of the two point counts at all large p would force equality of the virtual semisimplifications, contradicting the display. Hence $C(n, p)$ is not elementary.

For $n = 3$ there is no partition with four parts at all, and for $n = 4, 6$ every such partition is biquanimous: $(1, 1, 1, 1)$ for $n = 4$, and $(3, 1, 1, 1)$ and $(2, 2, 1, 1)$ for $n = 6$. $\square$

Theorem 6

ours — proved for every n

Proved, not measured: the witness $(n - 3, 1, 1, 1)$ is uniform in n , so no computation is being extrapolated. Independently checked against exhaustive enumeration of the length-four partitions for $n \leq 40$, and the witness itself for $n \leq 60$, in `P3c.py`. The weight-one identity it rests on needs Ghorpade–Lachaud [8, Prop. 3.2] for $H^1(V_\lambda)$, proper base change and Leray to carry it to the resolution, and Chebotarev with Brauer–Nesbitt at the end.

Remark 7. Two features of the argument deserve emphasis. First, no linear independence of newforms and no separation of isogeny classes is needed: isogenous curves *reinforce* one another in the displayed sum. Second, the same argument excludes more than the stated conclusion — it rules out polynomials in p with coefficients given by arbitrary Artin-type class functions of finite image, not only quadratic characters.

Remark 8. The threshold is $l = 4$ *exactly*, not $l \geq 4$. A non-biquanimous λ with $l \geq 5$ need not produce anything non-elementary: for $n = 5$ the partition (1^5) is not biquanimous, yet $V_{(1^5)}$ is the Clebsch surface, whose 27 lines are defined over $\mathbb{Q}(\sqrt{5})$ and whose count $p^2 + (6 + \chi_p(5))p + 1$ is elementary. Nor is the Clebsch exceptional: it is the member $a = 1$ of an infinite family $V_{(a, 1, 1, 1, 1)}$ whose count and whose line field $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$ are given in Proposition 11, and that field is abelian of exponent two for every a . At $n = 7$, however, $V_{(2, 2, 1, 1, 1)}$ has a *non-abelian* line field, containing the splitting field of a cubic of discriminant -140 and Galois group S_3 — Proposition 9, where that cubic is not quoted but produced by the same pencil. Two length-five strata with exactly two distinct weights each, and the line fields land on opposite sides of abelian: the arithmetic of this stratum is a separate and richer problem, and Theorem 6 does not require solving it.

The count behind the last sentence is worth writing down, because the same pencil that gives it is what produces the cubic, and because it is the smallest case in this note where the character attached to a surface is not quadratic.

The stratum tower: almost all the sieve mass sits on floors that carry no weight one; the floor that decides is the thin one

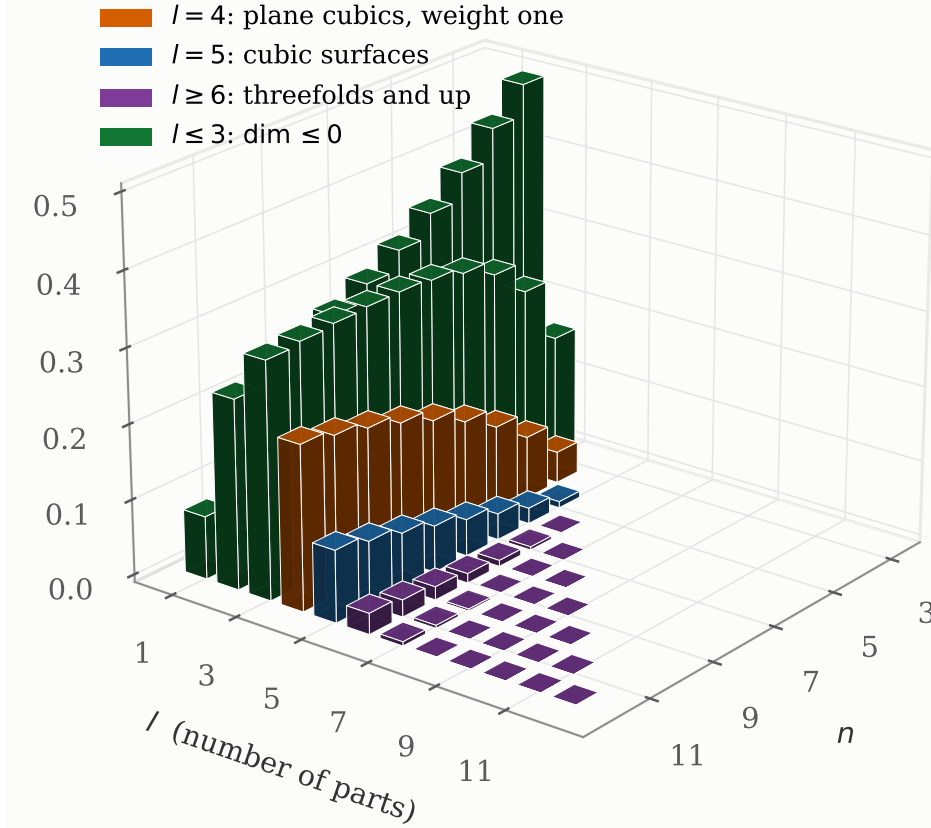


Figure 4: Why the argument is not delicate. Height is the sieve mass $\sum 1/z_\lambda$ of each stratum. Almost all of it sits on floors that carry no weight one — short strata by dimension, long ones by weak Lefschetz — and the floor that decides is the thin band in front.

Proposition 9 (the count of $V_{(2,2,1,1,1)}$). *Let $P(r) = r^3 + 8r^2 + 16r + 10$, irreducible over $\mathbb{Q}$, of discriminant -140 and Galois group S_3 , and let R_p be the number of distinct roots of P in $\mathbb{F}_p$. For every prime $p > 3$,*

$$\#V_{(2,2,1,1,1)}(\mathbb{F}_p) = p^2 + 1 + p \left(3 + \chi_p(5) + \sum_{\substack{r \in \mathbb{F}_p \\ P(r)=0}} \chi_p(-2r) \right),$$

the sum running over the distinct roots of P in $\mathbb{F}_p$.

Proof. Put $s = x_1 + x_2$, $u = x_1 - x_2$, $t = x_3 + x_4$, $v = x_3 - x_4$ and eliminate $x_5 = -(2s + t)$ by the linear equation; four times the cubic becomes

$$F = 2s^3 + 6su^2 + t^3 + 3tv^2 - 4(2s + t)^3$$

in $\mathbb{P}^3$ with coordinates $(s : u : t : v)$. The line $L : s = t = 0$ lies on F , and the plane $t = rs$ of the pencil through L meets the surface in L together with a residual conic, since $F|_{t=rs} = s(6u^2 + 3rv^2 - 3P(r)s^2)$; dividing by 3,

$$C_r : 2u^2 + rv^2 - P(r)s^2 = 0 \quad \text{in } (s : u : v).$$

Its discriminant is $-2rP(r)$, so the degenerate fibres are $r = 0$, the roots of P , and $r = \infty$; and $P(0) = 10$, so for $p > 5$ the first two do not meet. A smooth ternary conic over $\mathbb{F}_p$ has $p + 1$ points, and C_r meets L — the line $s = 0$ of that plane — in $1 + \chi_p(-2r)$ points, so a smooth fibre contributes $p - \chi_p(-2r)$. At $r = 0$ the form is $2u^2 - 10s^2$, a pair of lines rational exactly when 20 is a square, and it meets L once, contributing $p + p\chi_p(5)$. At a root of P the form is $2u^2 + rv^2$, a pair of lines rational exactly when $-2r$ is a square, contributing $p + (p - 1)\chi_p(-2r)$. The plane $s = 0$ meets the surface in $3t(v^2 - t^2) = 0$, three lines through one point of L , contributing $2p$. Adding $\#L = p + 1$ and using $\sum_{r \in \mathbb{F}_p} \chi_p(-2r) = 0$ to absorb the smooth fibres gives the identity. At $p = 5$ the two degenerate branches meet, since $P(0) = 10 \equiv 0$, and the bookkeeping is best done by hand rather than by the general count: the fibre $r = 0$ is a double line and contributes 5 points off L , the fibre $r = 1$ contributes 1, the fibres $r = 2, 3, 4$ contribute 4, 4, 6, the fibre at infinity contributes 10, and L contributes 6, for 36 in all, which is what the first displayed formula gives. Only $p = 3$ is excluded, where the pencil computation divides by 3; there both original equations degenerate to the same linear one and $V_{(2,2,1,1,1)}(\mathbb{F}_3)$ is all of $\mathbf{P}^3(\mathbb{F}_3)$. $\square$

Corollary 10 (the two arithmetic inputs separate). *For $p > 5$,*

$$\#V_{(2,2,1,1,1)}(\mathbb{F}_p) = p^2 + 1 + p(3 + \chi_p(5)(1 + R_p)) ,$$

so the count depends on exactly two things: whether 5 is a square mod p , and how many roots P has there.

Proof. Write $P(r) = r(r + 4)^2 + 10$. A root r in $\mathbb{F}_p$ has $r + 4 \neq 0$ for $p > 5$, and $r(r + 4)^2 = -10$ gives

$$-2r = \frac{20}{(r + 4)^2} = 5\left(\frac{2}{r + 4}\right)^2, \quad \text{so} \quad \chi_p(-2r) = \chi_p(5)$$

for *every* root at once, and the sum in Proposition 9 becomes $R_p \chi_p(5)$. At $p = 5$ this fails — $\chi_5(5) = 0$ while $\chi_5(-2) = -1$ — which is why the corollary starts at $p > 5$ while the proposition does not. $\square$

Proposition 9 and Corollary 10 — ours — a worked pencil and an identity, neither a new method

The method is the classical one for a cubic surface containing a line: project from the line and count the conic bundle fibre by fibre. Nothing in it is new. What the proposition supplies is the evaluation for this surface, and in particular the cubic P , which the remark above needs and which comes out of the pencil rather than being fitted to the counts. The corollary is two lines of algebra on top of it; its merit is only that it separates the two arithmetic inputs the character sum mixes, which is what makes the comparison with the abelian family of Proposition 11 legible. The S_3 here is the reason $V_{(2,2,1,1,1)}$ sits outside that family.

Measured: Proposition 9 against brute-force enumeration in the 33 primes $5 \leq p \leq 149$, with no exception, and $p = 3$ checked to be a genuine exception. The derivation's four algebraic steps are verified symbolically, not just their consequence; two negative controls — dropping $\chi_p(5)$, and $\chi_p(2r)$ for $\chi_p(-2r)$ — fail as they must. The collapsed form is checked separately, together with a negative control replacing 5 by another integer, which fails as it must; and $p = 5$ is confirmed to satisfy the first form and not the second. P splits completely at $p = 71, 73, 83, 149$ in that range, and $\chi_p(5)$ is $+1$ at 71 and 149 and -1 at 73 and 83 — so P splitting is not the same as the residual conics splitting into rational lines, and the two should not be conflated. See `gates/P3c_haz_conicas_22111.py` and `gates/P3c_superficie_22111.py`. The same pencil proves the abelian family of the remark, so that the comparison there rests on two proofs.

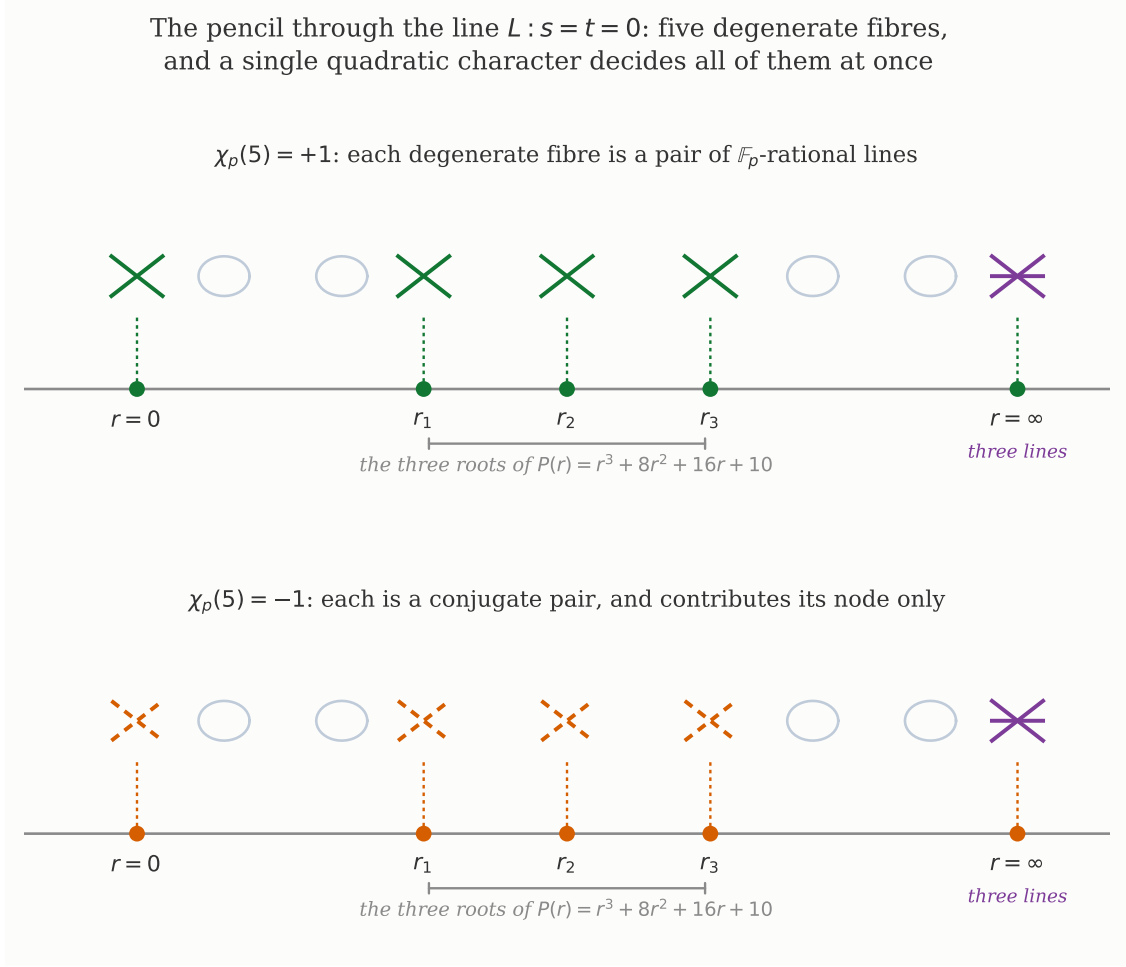


Figure 5: The proof of Proposition 9, drawn. The parameter line carries the pencil of planes through L ; almost every fibre is a smooth conic, which has $p + 1$ points whatever p is and therefore cannot carry arithmetic. Only five fibres degenerate — $r = 0$, the three roots of P , and $r = \infty$ — and by Corollary 10 *one* quadratic character decides four of them together: solid crosses when $\chi_p(5) = +1$, dashed conjugate pairs when it is -1 . The fibre at infinity is drawn in the third colour in both rows because it is three lines regardless, and it is the one contribution the character does not touch.

Proposition 11 (the family $V_{(a,1,1,1,1)}$). *Let a be a positive integer with $a \neq 2, 4$, and put $D_a = -3(a^2 - 4)$ and $E_a = (a^2 - 4)(a^2 - 16)$. For every prime $p \nmid 6a(a^2 - 4)(a^2 - 16)$,*

$$\#V_{(a,1,1,1,1)}(\mathbb{F}_p) = p^2 + [3 + 3\chi_p(D_a) + \chi_p(E_a)]p + 1,$$

and the twenty-seven lines are defined over $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$, an abelian extension of exponent two.

Proof. Eliminating the weight- a variable puts the surface in $\mathbf{P}^3$ as $a^2(x_1^3 + x_2^3 + x_3^3 + x_4^3) - (x_1 + x_2 + x_3 + x_4)^3 = 0$. With $s = x_1 + x_2$, $u = x_1 - x_2$, $t = x_3 + x_4$, $v = x_3 - x_4$ this is $a^2(s^3 + t^3 + 3su^2 + 3tv^2) - 4(s + t)^3 = 0$, the line $L : s = t = 0$ lies on it, and the plane $t = rs$ of the pencil through L meets the surface in L together with the residual conic

$$3a^2u^2 + 3a^2rv^2 + A_a(r)s^2 = 0, \quad A_a(r) = (r + 1)[(a^2 - 4)r^2 - (a^2 + 8)r + (a^2 - 4)].$$

There are five degenerate fibres — $r = 0$, $r = \infty$, $r = -1$ and the two roots

$$r_{\pm} = \frac{(a^2 + 8) \pm a\sqrt{3(16 - a^2)}}{2(a^2 - 4)}$$

of the quadratic factor. In the fibre $t = r_{\pm}s$ the conic is $3a^2(u^2 + r_{\pm}v^2) = 0$, the pair of lines $u = \pm\sqrt{-r_{\pm}}v$, and, for a suitable choice of the square roots, $\sqrt{-r_{\pm}} = (\sqrt{E_a} \pm a\sqrt{D_a})/2(a^2 - 4)$; these two numbers have product 1, sum $\sqrt{E_a}/(a^2 - 4)$ and difference $a\sqrt{D_a}/(a^2 - 4)$, so they generate exactly $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$, which also contains $r_{\pm}$. The fibres $r = 0$ and $r = \infty$ split into lines over $\mathbb{Q}(\sqrt{D_a})$ and the fibre $r = -1$ over $\mathbb{Q}$, so L and the ten lines meeting it are defined over $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$. The stabiliser of L in $W(E_6)$, isomorphic to $W(D_5)$, acts faithfully on those ten lines; the Galois group of the lines fixes L , so the subgroup acting trivially on the ten acts trivially on all twenty-seven, and the field of those ten is the field of the twenty-seven. Counting fibre by fibre as in Proposition 9, and using $\sum_r \chi_p(\cdot) = 0$ to absorb the smooth ones, gives the displayed identity whenever p divides none of 6, a , $a^2 - 4$, $a^2 - 16$. The excluded $a = 2, 4$ are exactly those for which D_a or E_a vanishes and the pencil degenerates. $\square$

Proposition 11

ours — a worked pencil, like its neighbour

The method is again the classical one for a cubic surface containing a line, and nothing in it is new. What the proposition supplies is the evaluation for this family and the field of its lines, which the discussion of the length-five stratum compares against Proposition 9: that comparison is only worth making if both sides are proved.

Measured: The change of variables, the line L , the residual conics and the degenerate fibres verified symbolically over $\mathbb{Q}[a]$, together with the discriminant identity behind the field; the root formula and the field $\mathbb{Q}(\sqrt{D_a}, \sqrt{E_a})$ checked in $\mathbb{Q}$ for $a \in \{1, 3, 5, 6, 7, 9\}$, and the fibre-by-fibre count over $\mathbb{F}_p$ for those a and $5 \leq p \leq 31$; 281 checks with no failure. The count against brute force on a grid of 120 pairs (a, p) with $a \leq 12$, $a \neq 2, 4$ and $5 \leq p \leq 43$: the formula fails in exactly the four cases $(5, 5)$, $(7, 7)$, $(10, 5)$, $(11, 11)$, which are exactly the pairs of the grid with $p \mid a$. Every failure therefore has $p \mid 6a(a^2 - 4)(a^2 - 16)$, so the hypothesis is sufficient; on the grid it is not necessary, since the only failures are those with $p \mid a$. See `gates/P3b_haz_conicas_Sa.sage` and `gates/P3c_familia_a11111.py`.

5 The four closed forms

Throughout χ_p is the quadratic character.

Proposition 12. *For $p > 3$ and $n \leq 4$, the condition $e_1 = e_3 = 0$ is equivalent to $T = -T$, and*

$$C(3, p) = \frac{p-1}{2}, \quad C(4, p) = \frac{(p-1)(p-3)}{8}.$$

Proof. Killing the coefficients of X^{n-1} and X^{n-3} leaves $X^3 + e_2X$ for $n = 3$, which is odd, and $X^4 + e_2X^2 + e_4$ for $n = 4$, which is even; in both cases the root set is stable under negation. Conversely $T = -T$ kills every odd moment. The counts are then the number of ways of choosing $\lfloor n/2 \rfloor$ of the $(p-1)/2$ antipodal pairs. $\square$

Proposition 12

elementary, and the two counts are folklore

For $n \leq 4$ the two conditions collapse to $T = -T$ and the count becomes a choice of antipodal pairs; both formulas are the kind of thing one rediscovers rather than proves, and we claim neither. They are here because the sieve needs every stratum, and because they fix the two boundary values the classification then has to reproduce.

λ	l	z_λ	V_λ	$\#V_\lambda(\mathbb{F}_p)$
(5)	1	5	—	$N_\lambda = 1$
(4, 1)	2	4	—	$N_\lambda = 1$
(3, 2)	2	6	—	$N_\lambda = 1$
(3, 1, 1)	3	6	binary form: linear $\times$ quadratic, disc = -15	$2 + \chi_p(-15)$
(2, 2, 1)	3	8	binary form: linear $\times$ quadratic, disc = 5	$2 + \chi_p(5)$
(2, 1, 1, 1)	4	12	smooth plane cubic, conductor 20	$p + 1 - a_p$
(1 ⁵)	5	120	Clebsch surface	$p^2 + (6 + \chi_p(5))p + 1$

Table 1: The seven strata of $n = 5$. Shade of z_λ is the sieve mass $1/z_\lambda$ the stratum carries, so the column is Figure 4 read as numbers. The tinted row is the one of length four, and it is the only stratum that is not elementary: it is the reason $C(5, p)$ is not elementary.

Theorem 13. For $p > 5$,

$$C(5, p) = \frac{p-1}{120} \left[p^2 - 4p + 61 + \chi_p(5)(p+15) + 20\chi_p(-15) + 10a_p \right],$$

where a_p is the Frobenius trace of the elliptic curve $E : Y^2 = X^3 + X^2 - X$ of conductor 20 and minimal discriminant 80.

Proof. The seven strata are listed in Table 1, and we justify them in turn. For $l \leq 2$ the system forces $x = 0$ and $N_\lambda = 1$. For $\lambda = (3, 1, 1)$ elimination gives the binary cubic $-3x(8x^2 + 9xy + 3y^2)$, whose quadratic factor has discriminant -15 , so $\#V = 2 + \chi_p(-15)$; for $\lambda = (2, 2, 1)$ it gives $-6(x+y)(x^2 + 3xy + y^2)$, of discriminant 5 , so $\#V = 2 + \chi_p(5)$. For $\lambda = (1^5)$ the variety is the Clebsch surface, with $\#V = p^2 + (6 + \chi_p(5))p + 1$.

The remaining stratum is $\lambda = (2, 1, 1, 1)$, the plane cubic $F(x, y, z) = 0$ with

$$F(x, y, z) = 2x(x + y + z)^2 + yz(y + z),$$

which is $-\frac{1}{3}$ times the cubic of (3) after eliminating x_4 , so it cuts out the same curve for $p > 3$. It is smooth, by Corollary 4 and because 5 is odd, and it carries the rational point $(0 : 1 : 0)$. The linear substitution

$$(U, V, W) = (z, 2x + 2y + z, 2x + z), \quad \det = -4,$$

satisfies the polynomial identity

$$V^2W - U^3 - U^2W + UW^2 = 4F(x, y, z),$$

so for $p > 2$ it is invertible and carries our cubic to $V^2W = U^3 + U^2W - UW^2$, that is to the affine model $v^2 = u^3 + u^2 - u$, which is E . Hence $\#V = p + 1 - a_p$. Substituting the seven values into (2) and simplifying gives the stated identity; the constant terms cancel and the rest assembles as displayed. $\square$

Theorem 14. For $p > 6$,

$$C(6, p) = \frac{p-1}{720} \left[p^3 - 9p^2 + 81p - 399 - 90\chi_p(-1) - 40\chi_p(-3) \right].$$

λ	l	z_λ	sign	V_λ	$\#V_\lambda(\mathbb{F}_p)$
(1^6)	6	720	+	Segre cubic, 10 nodes	$p^3 + 6p^2 - 4p + 1$
$(2, 1^4)$	5	48	−	Cayley cubic, 4 nodes	$p^2 + 3p + 1$
$(3, 1^3)$	4	18	+	nodal plane cubic, 1 node	$p + 1 - \chi_p(-3)$
$(2^2, 1^2)$	4	16	+	line + conic, 2 nodes	$2p$
$(4, 1^2)$	3	8	−	binary cubic, disc = −4	$2 + \chi_p(-1)$
$(3, 2, 1)$	3	6	−	binary cubic, a double root	2
(2^3)	3	48	−	three points	3
and for $l \leq 2$: $N_{(6)} = N_{(5,1)} = N_{(4,2)} = 1$, $N_{(3,3)} = p$.					

Table 2: The eleven strata of $n = 6$. Shade of z_λ is again the sieve mass. The two tinted rows are the length-four band, and here — unlike Table 1 — both are biquanamous, so both carry a node, neither is an elliptic curve, and the count stays elementary. This is Theorem 6 in one column.

Proof. The eleven strata are listed in Table 2. The first two entries are the Segre and Cayley cubics. Their counts can be read off the blow-up models — the Segre cubic is $\mathbb{P}^3$ blown up at five points with the ten joining lines contracted, the Cayley cubic has minimal resolution $\mathbb{P}^2$ blown up at six points with four (-2) -curves contracted — but that route needs the centres and the contractions to descend to $\mathbb{F}_p$, which is more than the rationality of the nodes gives for free. We therefore count directly, by pairing the variables; the blow-up models stay as the geometric reading, not as the proof.

For $\lambda = (1^6)$ set $(x_1, x_2) = \frac{1}{2}(s + u, s - u)$, $(x_3, x_4) = \frac{1}{2}(t + v, t - v)$ and $(x_5, x_6) = \frac{1}{2}(-s - t + w, -s - t - w)$. The linear equation holds identically and the cubic becomes

$$su^2 + tv^2 - (s + t)w^2 - st(s + t) = 0.$$

The plane $s = t = 0$ lies on it and contributes $p^2 + p + 1$. Off it, fix $(s : t) \in \mathbf{P}^1$. For $s \neq 0$ put $s = 1, t = r$; when $r \neq 0, -1$ the equation is $u^2 + rv^2 - (1 + r)w^2 = r(1 + r)$, which has $p^2 + p$ solutions, since $u^2 + rv^2 = b$ has $p - \chi_p(-r)$ solutions for $b \neq 0$ and $p + (p - 1)\chi_p(-r)$ for $b = 0$, and $w^2 = -r$ has $1 + \chi_p(-r)$ roots. The three remaining parameters $r = 0, r = -1$ and $(s : t) = (0 : 1)$ each give a difference of two squares with a free coordinate, hence $2p^2 - p$ solutions. Adding,

$$\#V_{(1^6)}(\mathbb{F}_p) = (p^2 + p + 1) + (p - 2)(p^2 + p) + 3(2p^2 - p) = p^3 + 6p^2 - 4p + 1.$$

For $\lambda = (2, 1^4)$ the same pairing of the four weight-one variables, with the weight-two one equal to $-(s + t)/2$, gives $su^2 + tv^2 - st(s + t) = 0$. Here $s = t = 0$ is a line L , and the plane $t = rs$ meets the surface in L and the conic $u^2 + rv^2 - r(r + 1)s^2 = 0$, smooth for $r \neq 0, -1$ and then contributing $p - \chi_p(-r)$ points off L ; $r = 0$ gives a double line with p , $r = -1$ a pair of rational lines with $2p - 1$, the plane at infinity p , and L itself $p + 1$. Since $\sum_{r \in \mathbb{F}_p} \chi_p(-r) = 0$,

$$\#V_{(2,1^4)}(\mathbb{F}_p) = (p^2 + p) + p + (p + 1) = p^2 + 3p + 1.$$

Both changes of variable are defined over $\mathbb{F}_p$ for $p > 3$ and divide only by 2. Substitution into (2) gives the identity. $\square$

Measured: Theorems 13 and 14 against the exact census in the 33 primes $7 \leq p \leq 151$, with no exception; and the traces a_p of E agreeing with the coefficients of $\eta(2z)^2\eta(10z)^2$, the newform of weight 2 and level 20, again with no exception. See `gates/P3b_B5_cerrada.py` and `gates/P3b_B6_cerrada.py`. That agreement over a finite range is a check and not a determination of the conductor; the conductor 20 and the minimal discriminant $80 = 2^4 \cdot 5$ of $y^2 = x^3 + x^2 - x$ are computed by Sage from the minimal model, which it obtains from the

plane cubic of $\lambda = (2, 1, 1, 1)$ by an explicit map over $\mathbb{Q}$, and the direct count of that cubic over $\mathbb{F}_p$ gives the same a_p in the same 33 primes. See `gates/P3b_B5_weierstrass.sage`.

Theorems 13 and 14

ours — identities, not fits

Each stratum is evaluated separately and the sum is assembled symbolically, so what is checked is an identity and not an agreement of numbers; the 33 primes are a control on top of it. The conductor is computed from the curve, not read from a database, and the eta product is an extra control that needs no computer algebra system.

Remark 15. The displayed expressions are valid in the stated ranges. If one insists on a single formula valid at every prime, the finitely many exceptions can be absorbed by terms of the form $1 - \chi_p(q^2)$, which equal 1 at $p = q$ and 0 elsewhere; being bounded, such terms do not affect Theorem 6.

6 What is open

We state these as problems.

The length-five stratum. For which *non-biquanimous* λ of length five — so that V_λ is a smooth cubic surface, and the twenty-seven lines are there to talk about — is the field of those lines abelian? The restriction is not a convenience: on a singular stratum the configuration is not twenty-seven lines at all, and the question as posed would have no subject. The family $V_{(a,1,1,1,1)}$ is abelian with group of exponent two, and $V_{(2,2,1,1,1)}$ is not; both have exactly two distinct weights, so the obvious invariant is not the answer. Note also that for the counts of Definition 5 the relevant condition is stronger than abelian: it is that the field be multiquadratic, since only characters of order dividing two can appear. The relevant machinery is that of rational pentahedral forms [3, 4] and the Frobenius traces of del Pezzo surfaces [1].

The general pattern $\{1, k\}$. What generalises is the *computation of the singular locus*, and only that: for the moment condition $M_1 = M_k = 0$ it consists of vectors of $(k-1)$ -st roots of unity with $\sum_i w_i \zeta_i = 0$, so the biquanimous condition is replaced by a vanishing weighted sum of roots of unity [11, 12]. The classification by weight one does *not* come along for the ride: the proof of Theorem 6 used that a singular plane cubic has rational components, and the paragraph on degree $d \geq 4$ above is exactly the reason that step is special to the exponent three. That generalisation is the subject of Part II [19]. One point is worth flagging here because it is easy to get wrong: it generalises the *moment* problem and not the coefficient problem — already at $k = n = 4$, $p = 11$, the set $T = \{0, 1, 2, 8\}$ has $e_1 = e_4 = 0$ but $M_4(T) \neq 0$.

Counting the obstructions. For even n , how many $\lambda \vdash n$ with $l \geq 4$ fail to be biquanimous? For $n = 8, 10, \dots, 34$ the counts are

$$1, 6, 9, 26, 39, 75, 109, 196, 248, 425, 557, 839, 1079, 1640.$$

The asymptotics of the unrestricted problem are in [5]; the restriction to $l \geq 4$ removes $O(n^2)$ partitions and so does not change the order.

Computations

All the computations are in the deposit `doi:10.5281/zenodo.22886540`: the program `P3c.py`, which ships with the note, and the scripts under `gates/` named in each *Measured* paragraph. `P3c.py` needs the standard library and `numpy`, and exits non-zero if any of its checks fails. It computes $C(n, p)$ by dynamic programming over (size, sum, sum of cubes), checked against direct enumeration for $n = 3, 4, 5$ and $p = 7, 11, 13$, and then checks, in the 11 primes $7 \leq p \leq 43$: the four closed forms of Proposition 12 and Theorems 13 and 14 against that census; the sieve (2), assembled from brute-force counts of the strata, against it for $n = 5, 6$; every row of Tables 1

and 2 against a brute-force count of its stratum; Proposition 9 and Corollary 10, with the failure of the collapsed form at $p = 5$; Proposition 11 for $a \leq 11$, $a \neq 2, 4$, under its hypothesis on p , with its failure at $a = p = 5$; and the identity of Proposition 16 at every point of $\mathbb{F}_p^4$, with the four nodes and $\det M$. Independently of the tables, it checks the sieve with every N_λ computed by dynamic programming over the two sums, against the census, for $3 \leq n \leq 7$ and every odd prime $p \leq 53$. On the combinatorial side it checks the count of Corollary 4 by subset-sum dynamic programming against a search over sign vectors for the 72 partitions with $l \geq 4$ and $n \leq 10$, the classification of Theorem 6 by exhaustive enumeration of the length-four partitions for $n \leq 40$, and the witness $(n - 3, 1, 1, 1)$ for $n \leq 60$. The singular locus itself, the symbolic steps of the pencils and of Proposition 16, and the conductor of E are not computed by P3c.py; they are in the scripts named alongside each statement.

A The objects in the tables, and where they come from

Tables 1 and 2 name three cubics. They are not three cubics that happen to have names: they are the three that nineteenth-century geometry singled out, and two of the statements below are about our strata rather than about people, so they are measured rather than recounted. Nothing here is needed to follow the proofs.

A.1 The two extremal cubics are the two longest strata of $n = 6$ (Table 2)

Over an algebraically closed field of characteristic zero, a cubic surface in $\mathbf{P}^3$ with only isolated double points can have at most four nodes, and there is exactly one that attains it: Cayley’s nodal cubic [21]. A cubic threefold in $\mathbf{P}^4$ can have at most ten, and again exactly one attains it: the Segre cubic [22]. Both bounds and both uniqueness statements are classical, and in characteristic zero; we use them only as names, since what the counts below need is Proposition 16, which is proved directly and carries its own hypothesis on the characteristic. Our sieve produces both, as the two longest strata of $n = 6$ — $\lambda = (1^6)$ gives Segre and $\lambda = (2, 1^4)$ gives Cayley — and the third surface in the tables, the stratum $\lambda = (1^5)$, the longest of $n = 5$, is the Clebsch diagonal surface.

Both uniqueness statements are *geometric*: they hold over $\bar{k}$ and up to projective equivalence, so on their own they do not say which model our strata are. Over $\mathbb{F}_p$ that distinction is not idle, since a twist has a different point count, and point counts are what these strata are here for. Proposition 16 settles it, and settles it by exhibiting the map rather than by invoking the classification.

Proposition 16 (the models are the standard ones, over the base field). *Let k be a field with $\text{char } k \nmid 6$.*

1. $V_{(1^6)}$ is the Segre cubic in its classical model $\{\sum_{i=1}^6 y_i = \sum_{i=1}^6 y_i^3 = 0\} \subset \mathbf{P}^5$ — the same scheme over $\mathbb{Z}$, not a form of it. Its ten nodes are the vectors with three entries $+1$ and three -1 , all k -rational.
2. Write $V_{(2,1^4)}$ in $\mathbf{P}^3$ by eliminating x_5 , with equation $F = 2x_1^3 + x_2^3 + x_3^3 + x_4^3 - (2x_1 + x_2 + x_3 + x_4)^3$. Its four nodes are

$$(1, -1, -1, -1), \quad (1, -1, -1, 1), \quad (1, -1, 1, -1), \quad (1, 1, -1, -1),$$

all k -rational and in general position. Let M be the matrix with those four columns, so $\det M = -8$. Then

$$F(Mu) = 24(u_0u_1u_2 + u_0u_1u_3 + u_0u_2u_3 + u_1u_2u_3),$$

and $V_{(2,1^4)}$ is projectively equivalent, over the prime field, to Cayley’s nodal cubic in its standard form.

Proof. (1) is a comparison of definitions, not a theorem: the classical model of the Segre cubic is the pair of equations displayed, which is (3) for $\lambda = (1^6)$. That the ten ± 1 vectors are singular is Lemma 2, or a direct check on the 2×6 Jacobian.

For (2), the nodes are obtained from $\partial F = 0$; they are four rational points, and the matrix of their coordinates has determinant -8 , so they are in general position and M lies in $\mathrm{GL}_4(\mathbb{Z}[1/2])$. In the coordinates u the four nodes are the coordinate points, and a cubic form singular at $(1 : 0 : 0 : 0)$ has no monomial divisible by u_0^2 ; imposing this at all four coordinate points leaves only $au_0u_1u_2 + bu_0u_1u_3 + cu_0u_2u_3 + du_1u_2u_3$. The displayed identity gives $a = b = c = d = 24$, so no rescaling of the variables is needed. The hypothesis on the characteristic is sharp and is exactly the content of the two constants: $\det M = -8$ degenerates when $\mathrm{char} k = 2$, and the factor $24 = 2^3 \cdot 3$ when $\mathrm{char} k$ is 2 or 3. $\square$

Both statements are over the base field, so the Segre and Cayley rows of Table 2 are counts on the standard models and transfer without twisting.

Proposition 16

ours — the map, not the classification

The classical statements are that four and ten nodes are the maxima and that one surface attains each; those are not ours and are not used. What is ours is the explicit identification over the base field: the four rational nodes, the matrix of their coordinates, and the identity $F(Mu) = 24(u_0u_1u_2 + \dots)$. That is what the point counts need, and it is checkable in one line of any computer algebra system without taking the classification on trust.

Measured: The four nodes of $V_{(2,1^4)}$ computed from the Jacobian rather than quoted; the identity $F(Mu) = 24(u_0u_1u_2 + \dots)$ verified symbolically; the ten nodes of $V_{(1^6)}$ verified to be singular points of the complete intersection; and a negative control in which the fourth column of M is replaced by $(1, 1, 1, 1)$, which destroys the Cayley form as it must. See `gates/P3c_cayley_segre_sobre_Q.py`. The extremal objects are not decoration here: they are two rows of Table 2, whose node counts 4 and 10 are what the blow-up models in the proof of Theorem 14 read geometrically, and the ten nodes of the Segre cubic are exactly the ten biquanamous splittings of (1^6) , which is Corollary 4 read backwards.

There is more than a coincidence of names. Classically, the Segre cubic cut by a hyperplane $y_i = y_j$ is Cayley’s nodal cubic surface. In the sieve, passing from (1^6) to $(2, 1^4)$ is *merging two parts of the partition* — and merging two parts is precisely setting $y_i = y_j$: substituting $y_1 = y_2 = x_1$ into $\sum_{i=1}^6 y_i = \sum_i y_i^3 = 0$ gives $2x_1 + x_2 + \dots + x_5 = 0$ and $2x_1^3 + x_2^3 + \dots + x_5^3 = 0$, which is $V_{(2,1^4)}$ on the nose. The combinatorial operation that organises the Li–Wan sieve and the geometric operation that relates the two extremal cubics are the same operation.

A.2 Where the $\sqrt{5}$ comes from (Table 1)

The Clebsch row carries the only quadratic character attached to a *smooth* surface in Table 1 — the singular strata contribute their own, and the assembled $C(5, p)$ carries $\chi_p(-15)$ as well — $\#V_{(1^5)}(\mathbb{F}_p) = p^2 + (6 + \chi_p(5))p + 1$, and the 5 in it is not a fitted constant. A smooth cubic surface contains twenty-seven lines; on the Clebsch surface fifteen of them are defined over $\mathbb{Q}$ and the remaining twelve are conjugate over $\mathbb{Q}(\sqrt{5})$. So over $\mathbb{F}_p$ one sees all twenty-seven lines when 5 is a quadratic residue and only fifteen when it is not, and the Frobenius trace on the Picard lattice, which determines the count [31], moves accordingly. The character in the count is the field of definition of the lines.

Measured: $V_{(1^6)}$ has exactly ten singular points for $p = 11, 13, 17, 19$, identified one by one as the ± 1 vectors summing to zero for $p = 13, 19, 23$; $V_{(2,1^4)}$ has exactly four for $p = 11, 13, 17, 19$; the hyperplane section $y_1 = y_2$ of $V_{(1^6)}$ equals $V_{(2,1^4)}$ point by point for $11 \leq p \leq 23$; $V_{(1^5)}$ is smooth for $11 \leq p \leq 29$; and the number of lines on $V_{(1^5)}(\mathbb{F}_p)$ is 27 for $p = 11, 19, 29, 31$ and 15 for $p = 7, 13, 17, 23$ — both branches of $\chi_p(5)$, by direct enumeration. See `gates/P3c_tres_cubicas.py`.

A.3 The twenty-seven lines were counted in a letter (Appendix A.2)

Cayley was the first to notice that a cubic surface carries only a definite, finite number of lines; Salmon showed that the number is twenty-seven. The theory was first worked out in correspondence between the two, and their results were then published independently, both in 1849 [25, §1.5.1]. The result that organises the whole subject, and that our Clebsch row silently uses, was therefore settled in letters: the correspondence is where it was found, not where it appeared.

Clebsch’s surface entered the world as an object one could hold. In 1872, at Göttingen, he presented a plaster model of it — a model of a cubic surface on which all twenty-seven lines are real and visible — and it later became part of Rodenberg’s series of cubic surface models. Clebsch died that same year, at thirty-nine.

Appendices A.3 to A.6 are historical framing and carry nothing mathematical; Appendices A.1 and A.2 are not, and carry their own proofs and measurements. The sources of the historical part are these: the attribution of A064914 is taken from the OEIS entry itself [23], and the 1995 date of A006827 from its own entry, the relation between the two Segres from [24], the Cayley–Salmon correspondence from [25, §1.5.1], the realisation of every subgroup class of $W(E_6)$ from [26, Thm. 0.1], and the two nineteenth-century dates of Figure 1 from the memoirs themselves [28, 29]. We repeat without a primary source the 1872 plaster model, its place in Rodenberg’s series and Clebsch’s age at his death, the later identification of Jordan’s group with $W(E_6)$, and Beniamino Segre’s 1955 theorem on ovals.

A.4 The word the classification turns on was contributed to a database (Section 4)

Theorem 6 turns on whether a partition can be split into two blocks of equal sum. The term for that is *biquanimous*, and it is not classical: it appears in the On-Line Encyclopedia of Integer Sequences as A064914 [23], contributed by Christian G. Bower on 12 October 2001, with the definition “a biquanimous partition is one that can be bisected into two equal sized parts: e.g. $3 + 2 + 1$ is a biquanimous partition of 6 as it contains 3 and $2 + 1$, but $5 + 1$ is not”. That entry counts *ordered* biquanimous partitions of $2n$, that is, compositions. The sequence our Definition refers to, the partitions of $2n$ that are *not* biquanimous, is A006827, which is older — it is in the 1995 Encyclopedia of Integer Sequences — and whose asymptotics are Erdős–Nicolas–Sárközy’s [5]. The name a stratum of ours is classified by first appears, as far as we can tell, on a database in 2001. A dated entry records a use of the word, not necessarily its first.

A.5 Why the twenty-seven lines have a Galois theory at all (Propositions 9 and 11)

Propositions 9 and 11 do not just count points: they land on two *fields*, one abelian of exponent two and one containing an S_3 . That such a question has an answer, and a finite one, is not obvious, and it has a name.

Jordan, in the *Traité des substitutions* of 1870 [27], computed the group of permutations of the twenty-seven lines that preserves their incidences. It has order 51840, and it was recognised much later as $W(E_6)$, the Weyl group of the exceptional root system: the twenty-seven lines are the weights of the 27-dimensional representation, and the incidence graph is the root system’s. So the Galois group of the lines of any smooth cubic surface over $\mathbb{Q}$ is a subgroup of $W(E_6)$, and asking for the field of the lines is asking which subgroup.

The question of which subgroups occur was settled by Elsenhans and Jahnel: *every* conjugacy class of subgroups of $W(E_6)$ is realised by a smooth cubic surface over $\mathbb{Q}$ [26, Thm. 0.1]. Our two strata are two of those classes, and the point of putting them side by side is that the weights

do not predict which: $V_{(a,1,1,1,1)}$ and $V_{(2,2,1,1,1)}$ both have exactly two distinct weights, and they fall on opposite sides of abelian. The open problem of Section 6 is which λ give which class — a question about $W(E_6)$ that the partition alone does not answer.

A.6 Two Segres (Appendix A.1)

The Segre of the Segre cubic is Corrado Segre, who studied it in 1887 [30]. The Segre who appears in the finite-geometry literature is Beniamino Segre, whose 1955 theorem that every oval in a Galois plane of odd order is a conic is usually credited with establishing that subject; the Vandermonde sets, against which Part II [19] checks its objects and finds that they are not of that kind, come from that tradition. The two Segres are not the same person, and they are not unrelated: Corrado was a cousin of Beniamino’s mother, and was his doctoral supervisor [24].

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